MARKO PAPUCKOVSKI YOUR NUTRITION NEEDS. 10 MARTIN STREET, ROSELANDS NSW 2196

Order ID 6463011
Lab ID 4203565
Patient ID P030929
Ext ID 26069-0152

MARK MARTIN

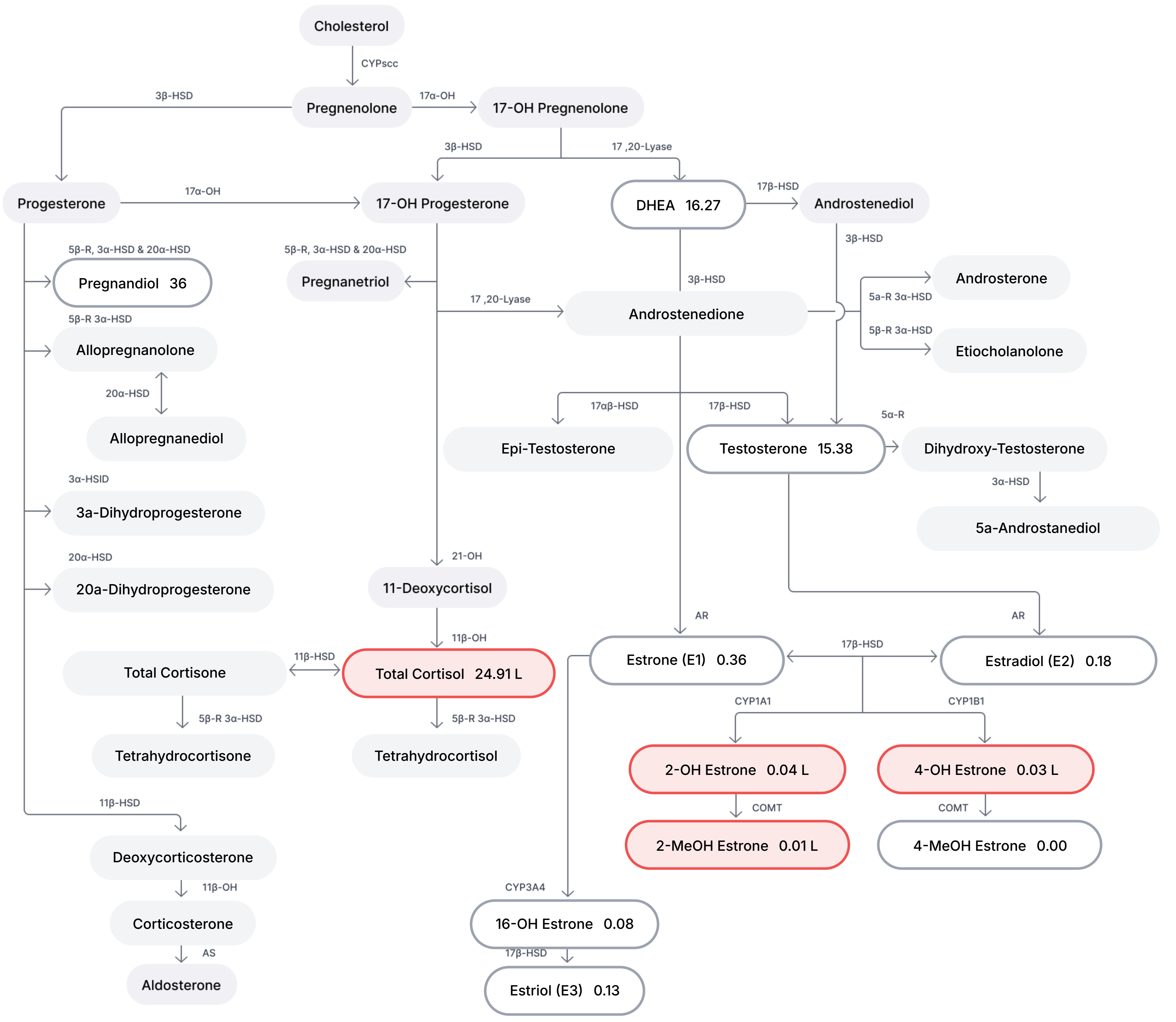
Sex: Male • 53yrs • 31-Jul-72
304/25 SWINBURNE STREET, LUTWYCHE QLD 4030

RECEIVED
10-Mar-26

EndoSCAN (Estrogen Metabolism)

Specimen type - Urine, Dried

Collected
07-Mar-26 05.00am



Legend Hormone not tested | Within range | Out of range | L = Low, LL = Critically Low H = High , HH = Critically High

Enzyme Abbreviations

5α-R	5a-Reductase	**3α-HSD**	3α-Hydroxysteroid dehydrogenase	**AR**	Aromatase
5β-R	5β-Reductase	**3β-HSD**	3β-Hydroxysteroid dehydrogenase	**AS**	Aldosterone Synthase
11β-oH	11β-Hydroxylase	**11β-HSD**	11β Hydroxysteroid dehydrogenase	**CYP**	Cytochrome p450 (scc, 1A1, 1B1 & 3A4)
17α-OH	17α-Hydroxylase	**17α-HSD**	17α-Hydroxysteroid dehydrogenase	**COMT**	Catechol-O-Methyl-Transferase
17,20-Lyase	Same enzyme as 17α-OH	**17β-HSD**	17β-Hydroxysteroid dehydrogenase		
21-OH	21-Hydroxylase	**20α-HSD**	20α-Hydroxysteroid dehydrogenase		

NATA No: 20770, Supervising Pathologists: Dr N. Maheshwari (Medical Director) & Dr M. Rana. Accreditation Scope available on governing accreditation body website.

NUTRIPATH • **PATIENT REPORT**

16 Harker St, Burwood VIC, 3125 • info@nutripath.com.au • 1300 688 522

NATA Accreditation: #20770

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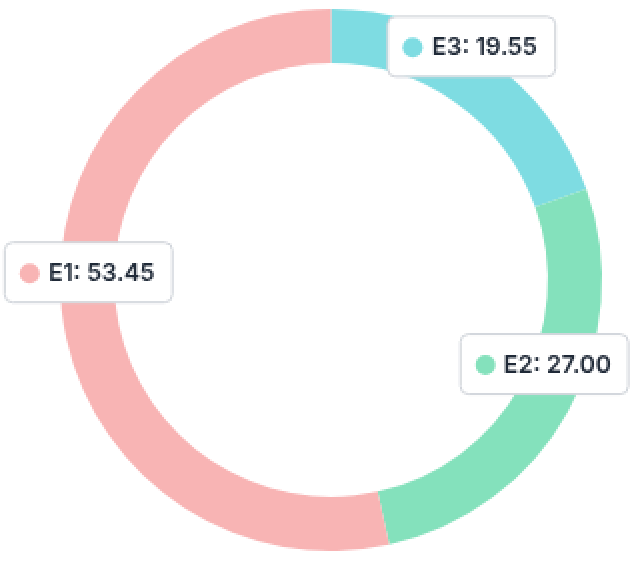
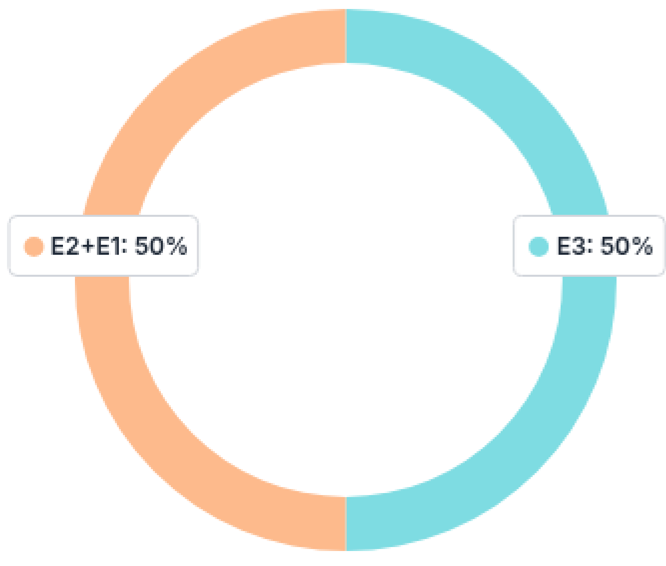
Primary Estrogens

TEST	RESULT	H/L	REFERENCE	UNITS
Estradiol (E2)	0.18		(0.10-0.44)	ug/gCR
Estrone (E1)	0.36		(0.30-1.40)	ug/gCR
Estriol (E3)	0.13		(0.10-0.62)	ug/gCR
Estrogen Quotient - E3/[E2+E1]	0.24	L	(>0.25)	ratio

Estrogens Balance (as %)



Healthy Estrogens Balance



OTHER PRIMARY HORMONES

TEST	RESULT	H/L	REFERENCE	UNITS
Pregnanediol	36		(25-150)	ug/gCR
Pregnanediol/Estradiol	196		(100-656)	ratio
Total 24hr Cortisol	24.91	L	(50.00-200.00)	ug/gCR
DHEA	16.27		(9.00-105.00)	ug/gCR
DHEA-S	494.5		(100.0-4000.0)	ug/gCR
Testosterone (TT)	15.38		(2.50-25.00)	ug/gCR

Urinary Creatinine

TEST	RESULT	H/L	REFERENCE	UNITS
Creatinine, Urine Pooled	3.36	H	(0.30-2.00)	mg/ml

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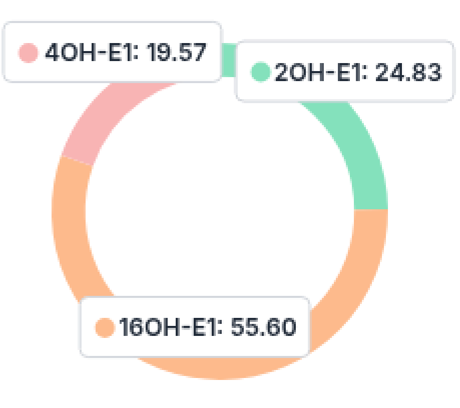
Estrogen Metabolism - Phase 1 (Hydroxylation)

TEST	RESULT	H/L	REFERENCE	UNITS
2-OH Estradiol	0.01		(0.01-0.39)	ug/gCR
2-OH Estrone	0.04	L	(0.10-1.88)	ug/gCR
4-OH Estradiol	<DL	L	(0.02-0.25)	ug/gCR
4-OH Estrone	0.03	L	(0.03-0.30)	ug/gCR
16-OH Estrone	0.08		(0.04-0.40)	ug/gCR
2-OH(E1+E2)/16-OHE1	0.60	L	(1.10-5.60)	ratio

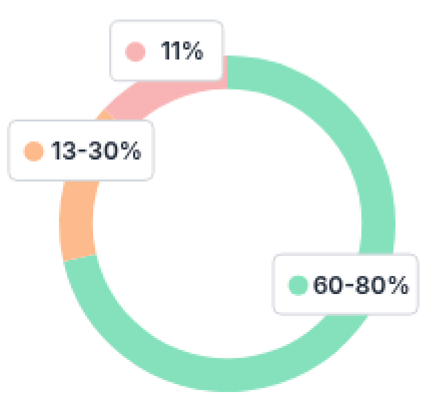
Estrogen Metabolism - Phase 2 (Methylation)

TEST	RESULT	H/L	REFERENCE	UNITS
2-MeOH Estradiol	0.01	L	(0.01-0.08)	ug/gCR
2-MeOH Estrone	0.01	L	(0.02-0.20)	ug/gCR
2-MeOH E1/2-OH E1	0.30		(0.05-0.50)	ratio
4-MeOH Estradiol	0.02		(<0.05)	ug/gCR
4-MeOH Estrone	<DL		(<0.05)	ug/gCR
4-MeOH E2/4-OH E2	4.39	H	(0.06-1.03)	ratio
4-MeOH E1/4-OH E1	0.07		(0.04-0.47)	ratio

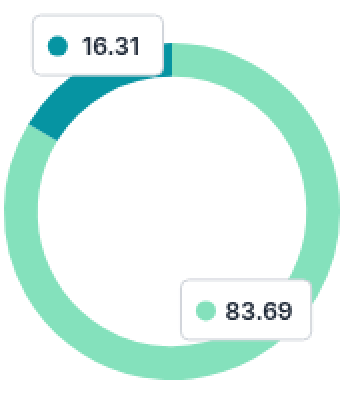
Metabolism Ph1 % (Hydroxylation)



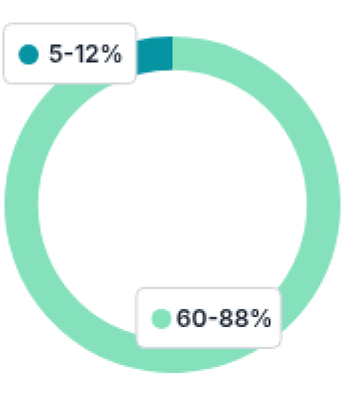
Healthy Ph1 Metabolism



Metabolism Ph2 % (Methylation)



Healthy Ph2 Metabolism



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ENDOCRINE DISRUPTORS

TEST	RESULT	H/L	REFERENCE	UNITS
Bisphenol A (BPA)	0.13		(<4.00)	ug/gCR
Polyfluoroalkyl Substances (PFAS)	0.02		(<0.70)	ug/gCR
Perfluorooctanoic Acid (PFOA)	0.00		(<0.10)	ug/gCR
Perfluorooctane Sulphonic Acid (PFOS)	0.02		(<0.60)	ug/gCR
Aluminium	<DL		(<14.00)	ug/gCR
Arsenic	<DL		(<26.50)	ug/gCR
Cadmium	<DL		(<0.60)	ug/gCR
Chromium	0.49		(<4.60)	ug/gCR
Lead	<DL		(<38.60)	ug/gCR
Mercury	<DL		(<17.9)	ug/gCR
Nickel	0.02		(<1.23)	ug/gCR

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Symptom Categories

Estrogen & Progesterone Deficiency	23.81%
Estrogen Dominance/Progesterone Deficiency	33.33%
Low Androgens	31.11%
High Androgens	33.33%
Low Cortisol	28.57%
High Cortisol	29.82%
Hypometabolism	29.17%
Metabolic Syndrome	36.67%

Symptom Score

1. MILD	2. MODERATE	3. SEVERE
Elevated triglycerides	Decreased libido	Ringing in ears
Swelling or puffy eyes/face	Decreased stamina	Hearing loss
Rapid heartbeat	Decreased erections	Difficulty sleeping
Low blood sugar	High blood pressure	Weight gain - Waist
Neck or back pain	Burned out feeling	
Increased urinary urge	Decreased mental sharpness	
Stress	Mental fatigue	
Nervous	Morning fatigue	
Heart palpitations	Weight gain - Breasts/hips	
High cholesterol	Evening fatigue	
Irritable		
Dizzy spells		

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Urinary Estrogens Comment

ESTROGEN QUOTIENT LOW:

The ratio of estriol (E3) to estrone (E1) and estradiol (E2) provides insights into estrogen metabolism. A low E3/(E1+E2) ratio may suggest an imbalance in estrogen metabolism, favoring the more potent estrogens (E1 and E2), potentially increasing the risk of estrogen-dominant conditions. Low levels of estriol relative to other estrogens may also indicate a disrupted metabolic pathway or insufficient estriol production, which could impact tissue health, mood regulation, and cardiovascular function.

2-HYDROXY-ESTRONE LOW:

2-Hydroxyestrone is a metabolite that suggests favorable detoxification of estrogen. Low levels of 2-OH estrone may indicate impaired phase I detoxification processes or liver dysfunction. This can lead to an accumulation of more potent, potentially harmful estrogen metabolites, increasing the risk of estrogen-related cancers and other health issues. Monitoring this marker is essential for assessing the body’s ability to detoxify estrogen and prevent excessive estrogenic activity.

4-HYDROXY-ESTRADIOL LOW:

4-Hydroxyestradiol is a genotoxic estrogen metabolite linked to oxidative stress and DNA damage. Low levels of 4-OH estradiol may reflect a well-regulated estrogen metabolism pathway, which is beneficial for minimizing oxidative damage. This can indicate a lower risk of estrogen-induced cancer and other oxidative stress-related conditions. A balanced production of estrogen metabolites is essential for overall health and minimizing risks associated with high estrogenic activity.

4-HYDROXY-ESTRONE LOW:

4-Hydroxyestrone, like 4-OH estradiol, is a genotoxic estrogen metabolite that can lead to oxidative damage and an increased risk of cancer. Low levels of 4-OH estrone may suggest efficient detoxification and a reduced risk of DNA damage. This could reflect a healthy metabolic pathway that minimizes oxidative stress and its associated risks.

2-HYDROXY-ESTROGENS/16-HYDROXY-ESTROGENS RATIO LOW:

The ratio of 2-OH estrone (E1 + E2) to 16α-OH estrone reflects estrogen metabolism balance. A low ratio may indicate a preference for the production of more potent estrogen metabolites, increasing the risk of estrogen-related diseases, including hormone-sensitive cancers. Low ratios may also point to reduced detoxification of estrogen and an increased overall estrogenic effect on the body.

2-METHOXY-ESTRADIOL LOW:

2-Methoxyestradiol (2-MeO E2) is a metabolite of estradiol with protective antioxidant properties. Low levels of 2-MeO estradiol suggest a reduced capacity for detoxifying estrogen metabolites and may indicate an increased risk of oxidative stress and estrogen-related damage. This could impact long-term health, especially in relation to cancer prevention and tissue health.

2-METHOXY-ESTRONE LOW:

2-Methoxyestrone is a metabolite that suggests efficient detoxification of estrogen. Low levels of 2-MeO estrone may signal poor estrogen clearance and a reduced capacity for detoxifying harmful estrogen metabolites. This could increase the risk of estrogen-related conditions, such as hormone-sensitive cancers, and indicate the need for enhanced liver function or metabolic support.

4-METHOXY-ESTRADIOL/4-HYDROXY-ESTRADIOL RATIO ELEVATED:

A high ratio of 4-MeO estradiol to 4-OH estradiol reflects enhanced detoxification and a reduced risk of oxidative damage and cancer. Elevated levels are generally beneficial, promoting the clearance of harmful metabolites and reducing the risk of estrogen-induced damage.

Urine Cortisol Comment

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TOTAL CORTISOL LOW:

Low total urinary cortisol suggests reduced adrenal glucocorticoid output, commonly seen in chronic HPA axis suppression, adrenal insufficiency, or prolonged physiological stress leading to "adrenal fatigue." This state may result in fatigue, hypotension, low mood, and impaired stress tolerance. Low cortisol also reduces the inhibitory effect on gonadotropins, which can paradoxically elevate DHEA or sex hormones initially, before a global hormonal decline occurs. Support includes restoring circadian rhythm, adaptogens, vitamin C, and B5, and assessing for adrenal or pituitary dysfunction.

Methodology

Liquid Chromatography-Mass Spectrometry (LC-MS/MS/MS), Inductively Coupled Plasma Mass Spectrometry (ICP-MS)